Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/11/2006
PI Name:	Drysdale, Alan E Ph.D.		
Project Title:	Equivalent System Mass Impacts of Clothing on Wastes and Mission Cost		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	alan.e.drysdale@boeing.com	**Fax:**	FY 321-383-2802
PI Organization Type:	NASA CENTER	**Phone:**	321-383-2857
Organization Name:	Boeing / NASA Kennedy Space Center		
PI Address 1:	PO Box 21233		
PI Address 2:			
PI Web Page:			
City:	KSC	**State:**	FL
Zip Code:	32815	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	03/03/2003	**End Date:**	03/02/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barker, Roger (NC State University) Thompson, Don (NC State University) Scruggs, Barbara (NC State University) Fratto, Lisa (NC State University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Resupply of clothing comprises about 16% - 19% of the mission equivalent system mass (ESM) for long duration missions. Disposing of the clothing as waste also constitutes a significant impact. Two approaches have been identified that show promise for large reductions in these impacts. · Mass of clothes can be reduced by using high-tech materials, which also provide other comfort and safety benefits. · Mass of clothes can be reduced by using a cleaning system and reusing clothes. This does require additional equipment and interfaces, but preliminary analyses show promise that the net ESM would be greatly reduced. For the International Space Station (ISS), crew clothing is supplied at a rate of about 194 kg and 1 m3 per 90-day increment for a crew of 3, though an attempt to reduce this is underway. The clothing, once worn, then becomes waste that must be managed. Crew debriefs (Increment 1) documented that odors produced from soiled clothing were more unpleasant for the crew than were odors from food trash. We propose to evaluate standards and requirements for providing clothing for human spaceflight missions, using ISS as a baseline and modifying and expanding the requirements for future missions. We will address two possible ways to reduce the mission impact of clothing; a redesign of the wardrobe, and cleaning and re-use of clothing in space. We will identify life-support related impacts that clothing has on subsystems and interfaces, options for processing the clothing, costs and benefits, and recommend particular approaches for particular near-term missions. Reduce clothing system ESM. The clothing system is one of the few areas where significant improvements in the ALS (advanced life support) project metric can be made.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential cross utilization of developed technology. In particular, reduced water and energy use, and lower vibration and noise, are likely.
Task Progress:	Current and historical clothing system practice for space missions has been investigated. Terrestrial clothing system approaches have been evaluated for applicability. Future missions and clothing system options for those missions have been assessed. For all missions, including the crew exploration vehicle (CEV) and the Lunar surface access module (LSAM), there is an opportunity to reduce equivalent system mass (ESM) by changing the clothing fabrics used. Modern fabrics have generally higher performance than traditional fabrics like cotton typically used on STS (shuttle) and ISS, in terms of mass, volume, useful life, water content, comfort, fire resistance, and fiber production. They can also allow clothing to be worn longer, reducing the rate of use. For vehicles with high volume equivalency, like the CEV, ESM can be reduced significantly by compressing the stowed clothing, both unused and used. There are some problems with this, particularly for long-duration missions, but modern fabrics would reduce these problems, and the benefits for these missions far outweigh the problems. A reduction in ESM of at least a factor of two is expected to be achieved by these two changes (in fabrics and reuse) while improving acceptability and safety. However, it is recognized that for short missions with small crews even a large percentage improvement in clothing system ESM is rather small in its effect on the overall mission. For long-duration missions such as a Lunar outpost, clothing system ESM is a large factor in mission cost. Clothes washing and drying is expected to be cost effective for these missions, reducing ESM by a factor of two for missions having a duration of the order of three months. For Mars surface missions lasting on the order of 600 days, much greater benefits are expected. Washer/dryer assessment was mostly based on a new design using largely-closed air and water loops to minimize ESM and impacts on the crew environment. The basic approach followed commercial-off-the-shelf (COTS) practice, and is expected to be achievable by modifying COTS equipment, including COTS water purification approaches. Only minor changes to COTS designs are needed for operation in fractional gravity. Significant changes would be required to operate in weightlessness , but these changes do seem feasible, and some of them would also benefit operation in partial gravity. None of the identified changes for operation in weightlessness would necessarily hinder operation in fractional gravity. Thus, a single design with perhaps minor software changes does seem feasible for both weightlessness and fractional gravity. However, fractional gravity functionality would be needed several years earlier, and is easier to develop and so would provide a more gradual developmental pathway, and this might be preferred. Development of a washer/dryer would require testing in weightlessness and fractional gravity. Some work will need to be done in parabolic flight, which can simulate both weightlessness and fractional gravity, but complete validation will require longer exposure to these conditions. Some work could be done using computational fluid dynamics, but the system is complex enough that it might be better not to rely on that. A washer/dryer would be cost effective for International Space Station (ISS), despite the high cost of power, cooling, and water regeneration, due to the long mission duration. One was not incorporated in the ISS design primarily because of development cost and risk. However, a washer/dryer would be more important for a long Lunar or Mars mission due to the expected high extravehicular activity (EVA) rates and increased delivery costs (location factors). Thus, development on ISS is justified by the need for clothes washing for the longer exploration missions, but could secondarily reduce ISS logistics loads. A washer/dryer was also found to be cost effective for the longest Lunar missions evaluated – a 98-day Lunar outpost. The benefit of washing compared to shipped clothing was calculated to be a factor of two. If the base was occupied for multiple missions, the benefit would be greater. A washer might also be particularly beneficial for both Lunar and Mars surfaces because of the pervasive dust found at both locations. Dust control is likely to be both mandatory and difficult, and there may be differences between the two locations. Thus, a demonstration unit should be flown both on ISS and on the Moon, to validate the design both in weightlessness

	and in fractional gravity as well as to reduce operating costs at these locations. A washer/dryer could be incorporated in a volume equivalent to about half an ISS rack, smaller if subscale or if multiple loads were washed per day. These two changes, in clothing materials and in reuse of clothing, are largely independent, and should both be pursued to reduce exploration mission costs. Because they can be implemented independently, they would have multiplicative effects. Use of modern clothing would reduce ESM, and also reduce the rate at which clothes would need to be washed, reducing the required size of washing/drying equipment.
Bibliography Type:	Description: (Last Updated: 04/14/2021)
Papers from Meeting Proceedings	Drysdale AE. "Recommendations for Clothing Systems for Advanced Missions." International Conference on Environmental Systems, Norfolk, Va, July 2006. International Conference on Environmental Systems, Norfolk, Va, July 2006. ICES paper 2006-01-2248. , Jul-2006